In [1]:
```python
# importing major libraries

import numpy as np
import pandas as pd

df = pd.read_csv('covid.csv')
```

In [2]:
```python
df.head()
```

Out[2]:

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes
2	42	Male	101.0	Mild	Delhi	No
3	31	Female	98.0	Mild	Kolkata	No
4	65	Female	101.0	Mild	Mumbai	No

In [3]:
```python
df.isnull().sum()
df.isnull().mean()
```

Out[3]:
```
age          0.0
gender       0.0
fever        0.1
cough        0.0
city         0.0
has_covid    0.0
dtype: float64
```

In [4]:
```python
df.shape
```

Out[4]:
```
(100, 6)
```

In [5]:
```python
from sklearn.model_selection import train_test_split
x = df.iloc[:,0:-1]
y = df.has_covid
```

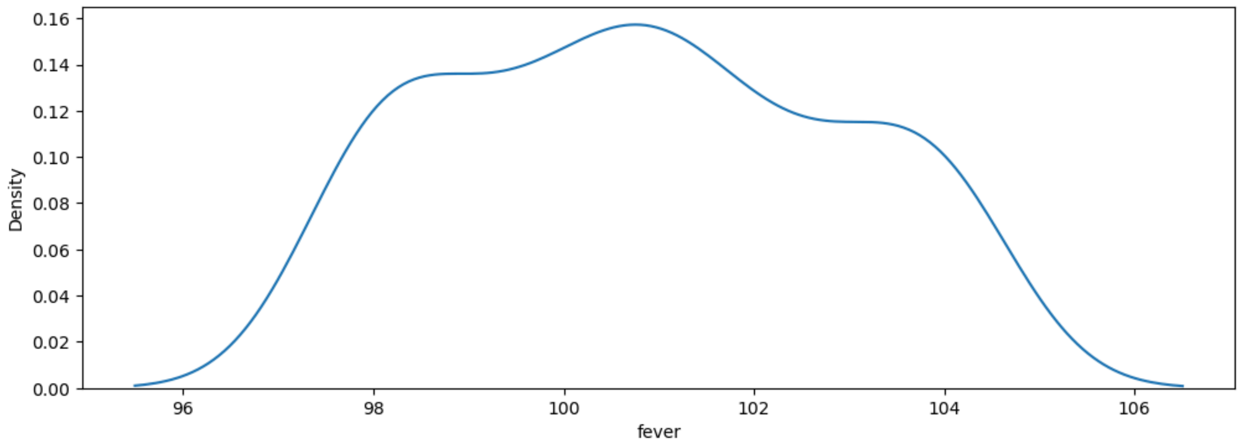
In [6]:
```python
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [7]:
```python
x_train.isnull().sum()
```

Out[7]:
```
age       0
gender    0
fever     9
cough     0
city      0
dtype: int64
```

In [8]:
```python
# imputers
import seaborn as sns
```

```python
import matplotlib.pyplot as plt
plt.figure(figsize=(12,4))
sns.kdeplot(df.fever)
plt.show()
```



In [9]:
```python
from sklearn.impute import SimpleImputer

si = SimpleImputer(strategy='mean')

# strategy, mean , median, constant, most_frequent(mode)
```

In [10]:
```python
si
```

Out[10]:
```
▾ SimpleImputer
SimpleImputer()
```

In [11]:
```python
x_train_fever = si.fit_transform(x_train[['fever']])
x_test_fever = si.transform(x_test[['fever']])
```

In [12]:
```python
# scaling

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
sc
```

Out[12]:
```
▾ StandardScaler
StandardScaler()
```

In [13]:
```python
# scaling
sc.fit(x_train[['age','fever']])
x_train_sc = pd.DataFrame(sc.transform(x_train[['age','fever']]),columns=sc.ge
x_test_sc = pd.DataFrame(sc.transform(x_test[['age','fever']]),columns=sc.get_
```

```
# sc.transform(x_test[['age','fever']])
```

In [14]:
```
# encoding
```

In [15]:
```
df.cough.unique()
# ordinal encoding

from sklearn.preprocessing import OrdinalEncoder
loe = ['Mild', 'Strong']
oe = OrdinalEncoder(categories=[loe],dtype=int)
oe
```

Out[15]:

```
▾ OrdinalEncoder
OrdinalEncoder(categories=[['Mild', 'Strong']], dtype=<class 'int'>)
```

In [16]:
```
oe.fit(x_train[['cough']])
```

Out[16]:

```
▾ OrdinalEncoder
OrdinalEncoder(categories=[['Mild', 'Strong']], dtype=<class 'int'>)
```

In [17]:
```
# transformation

x_train_cough = pd.DataFrame(oe.transform(x_train[['cough']]),columns=oe.get_f
x_test_cough = pd.DataFrame(oe.transform(x_test[['cough']]),columns=oe.get_fea
```

In [18]:
```
# df
df.city.unique()
```

Out[18]:
```
array(['Kolkata', 'Delhi', 'Mumbai', 'Bangalore'], dtype=object)
```

In [19]:
```
from sklearn.preprocessing import OneHotEncoder

ohecity = OneHotEncoder(sparse_output=False)
gender = ['Female','Male']
ohegender = OneHotEncoder(categories=[gender],sparse_output=False,drop='first'
```

In [20]:
```
ohecity.fit(x_train[['city']])
```

Out[20]:

```
▾ OneHotEncoder
OneHotEncoder(sparse_output=False)
```

In [21]:
```
x_train_city = pd.DataFrame(ohecity.transform(x_train[['city']]),columns=oheci
x_test_city = pd.DataFrame(ohecity.transform(x_test[['city']]),columns=ohecity
```

In [22]:
```
ohegender.fit(x_train[['gender']])
```

Out[22]:

OneHotEncoder

```
OneHotEncoder(categories=[['Female', 'Male']], drop='first',
              sparse_output=False)
```

In [23]:

```
x_train_gender = pd.DataFrame(ohegender.transform(x_train[['gender']]),columns
x_test_gender = pd.DataFrame(ohegender.transform(x_test[['gender']]),columns=c
```

In [24]:

```
x_train_gender
x_train_city
x_train_cough
x_train_fever
x_train_sc
```

Out[24]:

	age	fever
0	1.566141	0.000000
1	-1.558945	-0.491423
2	-0.983271	-0.491423
3	-0.654315	-0.491423
4	1.237185	0.982846
...	...	...
75	-0.777674	0.491423
76	1.319424	1.474269
77	0.332554	1.474269
78	1.607261	0.491423
79	-1.312228	-0.491423

80 rows × 2 columns

In [ ]: